A Unified Pipeline for Calculating Polygenic Risk Scores for Dementia and Psychiatric Disorders in the Dementias Platform UK

Alieyeh Sarabandi Moghaddam, Ammar Shahabuddin, Ganna Leoneko, Emma Squires, John Gallacher, Valentina Escott-Price, Sarah Bauermeister

Swansea University Prifysgol Abertawe

Dementias Platform UK | Data Portal

University of Oxford

Introduction

Dementias Platform UK (DPUK) is a national research infrastructure integrating several cohorts, providing secure access to multimodal data for dementia research. While genomic data holds great potential, its use in dementia has been hindered by computational bottlenecks and a narrow focus.

Accessibility: High technical barriers limited use by non-geneticists.

Computational Bottleneck: Sequential genotype imputation is slow, taking weeks per cohort

Limited Scope: focusing only on dementia and not including the established genetic and clinical links with psychiatric disorders can lead to missing important links.

Our Solution

We developed a unified, scalable pipeline for genomic quality control (QC), imputation, and PRS calculation, broadening the scope to include psychiatric disorders and dramatically improving efficiency.

Cohorts Used

- Utilized 6 DPUK cohorts: Brains for Dementia Research (BDR), Generation Scotland (GS), Epic Norfolk (EN), National Survey of Health and Development (NSHD), Airwaves 1 & 2 (AW1, AW2).
- Combined data from 63,887 individuals, primarily of European ancestry.

Cohort	Age range	Sample size	Variants	Chip
Brains for Dementia Research (BDR)	65+	570	478633	NeuroChip
Generation Scotland (GS)	18+	20032	604858	IlluminaExomeArray
Epic Norfolk (EN)	40+	21041	728244	Axiom 2020
National Survey of Health and Development (NSHD)	53	2864	11081207	NeuroX2
Airwaves 1 (AW1)	18+	4493	845487	Affymetrix
Airwaves 2 (AW2)	18+	14887	542667	IlluminacoreExome

Pipeline Workflow

The current, optimized pipeline follows the process of quality control, efficient imputation, and comprehensive PRS calculation within the DPUK Trusted Research Environment.

Quality Control (QC) & Pre-Processing

PLINK1.9 and R are used for this stage. Sample-level QC is conducted, with exclusion based on high genotype missingness (>5%), heterozygosity outliers, sex discrepancies, and cryptic relatedness. The, variant-level QC is done with exclusion of SNPs with missingness >5%, minor allele frequency (MAF) <1%, and deviation from Hardy-Weinberg equilibrium ($p < 1\times10^{-6}$). Principal Component Analysis (PCA) is then performed against the 1000 Genomes reference panel to control for population stratification.

Parallelized Phasing, Imputation and Post-Imputation QC

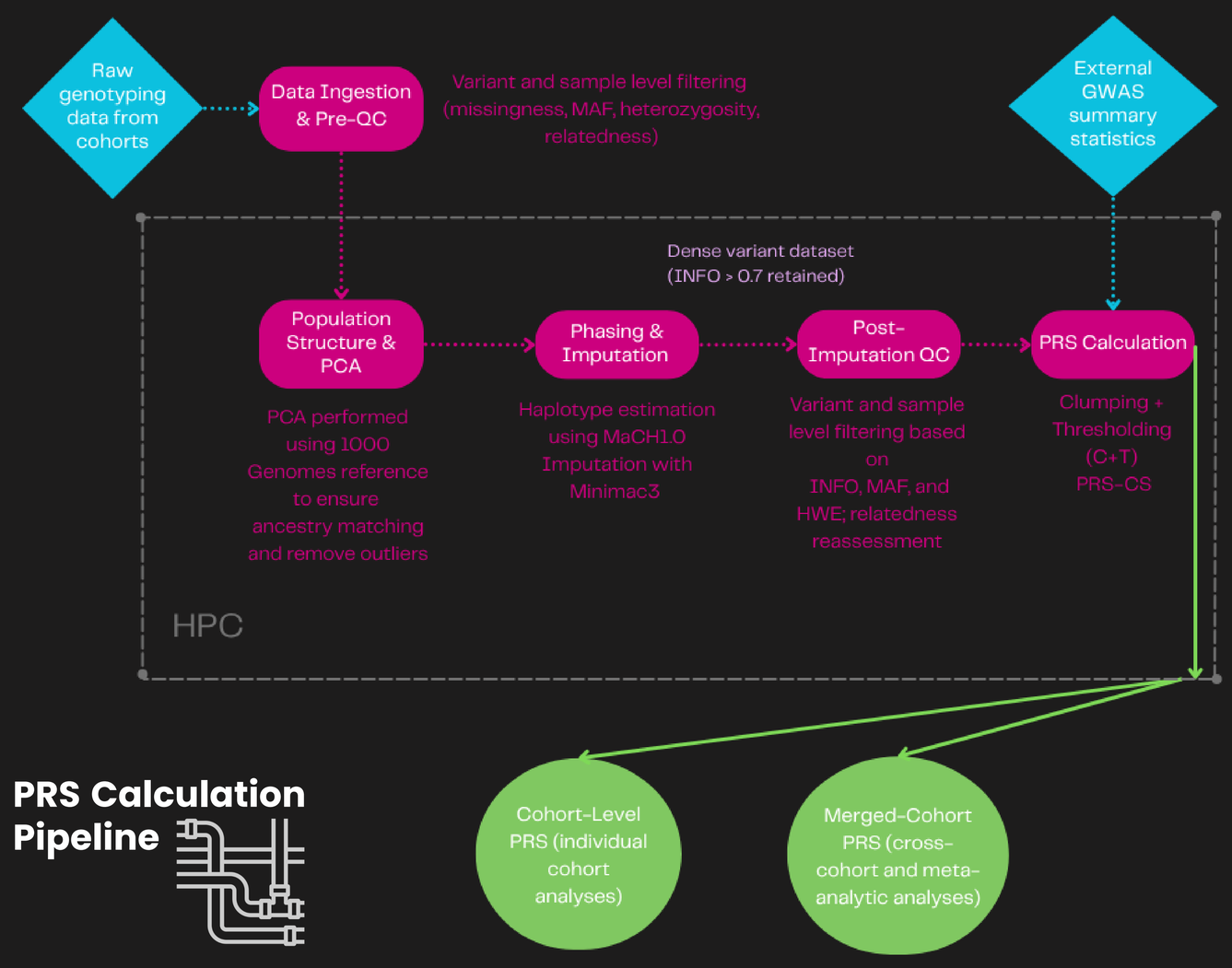
To maximize efficiency, the pipeline employs a parallelized approach for the computationally intensive imputation step. For the phasing stage, Haplotypes are estimated using MaCH1.0. Genotypes are then imputed using Minimac3 and the 1000 Genomes reference panel. The process is executed using SLURM job arrays to impute all chromosomes in parallel, drastically reducing runtime from weeks to hours. Variants are then filtered with similar criteria as the first QC step, as well as based on imputation quality (INFO score >0.7) and relatedness.

Polygenic Risk Score (PRS) Calculation

The pipeline generates PRS for a wide range of disorders to facilitate cross-phenotype research.
PRS are calculated for both neurodegenerative dementias (Alzheimer's, Parkinson's, FTD, ALS, Lewy Body) and psychiatric disorders (Schizophrenia, Bipolar Disorder, Depression, Anxiety).
Two primary methods are supported:

1. Clumping & Thresholding (C+T): Identifies independent SNPs via LD clumping (R^2 <0.1) and tests multiple p-value thresholds.
2. PRS-CS: A Bayesian method using continuous shrinkage priors, which improves predictive accuracy by incorporating more genetic information.

PRS are adjusted for principal components and standardized against the 1000 Genomes European population for comparability across cohorts. Research-ready imputed genetic data and pre-calculated PRS are available within the secure DPUK Data Portal. Researchers can use these derived datasets directly or leverage the adjustable pipeline for custom analyses.



PRS Calculation Pipeline

Strengths and Impacts

High Efficiency: Parallelized imputation overcomes the major computational bottleneck, accelerating genomic analysis.

Broadened Scope: First DPUK genomics pipeline to include PRS for major psychiatric disorders, enabling research into shared aetiology.

Enhanced Accessibility: Provides pre-computed PRS and imputed data within the TRE, allowing researchers to bypass many intensive computational steps, therefore democratizing access to complex genetic analysis for the wider research community.

Interoperability & Flexibility: A Well documented, adjustable pipeline which caters to diverse research needs, from single-cohort to cross-cohort analyses.

Future Work

Integrating diverse, non-European ancestry cohorts.

Validating the PRS scores against longitudinal cognitive decline and biomarker data.

Incorporating multi-omics such as, transcriptomic and proteomic data for a more holistic view.